CS 532 HW1

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Chapter 1

R3

Q: Why are standards important for protocols?
A: They ensure interoperability across vendors, predictable behavior, scalability, and security. They reduce ambiguity and enable widespread adoption and backward compatibility.

R14

Q: Why will two ISPs at the same level of the hierarchy often peer with each other? How does an IXP earn money?
A: To avoid transit fees, reduce latency, and balance traffic. IXPs charge port fees, memberships, cross-connects, and colocation services.

R16

Q: Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?
A: Processing, queueing, transmission, propagation (per hop). Constants: transmission, propagation, and typical processing. Variable: queueing (load-dependent); processing can vary slightly under load.

R23

Q: What are the five layers in the Internet protocol stack? What are the principalresponsibilities of each of these layers?
A: Application: network services (HTTP/DNS). Transport: end-to-end, reliability, multiplexing (TCP/UDP). Network: addressing/routing (IP). Link: framing/MAC/error detection (Ethernet/Wi-Fi). Physical: bits/signaling over medium.

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Q: Consider the queuing delay in a router buffer. Let I denote traffic intensity; that is, $I = \frac{La}{R}$. Suppose that the queuing delay takes the form $\frac{IL}{R(1-I)}$ for $I < 1$.

1. Provide a formula for the total delay, that is, the queuing delay plus the transmission delay.
   Given queueing delay $d_q = \frac{IL}{R(1-I)}$ and transmission delay $\frac{L}{R}$

$$D = d_q + d_t = \frac{IL}{R(1-I)} + \frac{L}{R} = \frac{IL + L - IL}{R(1-I)} = \frac{L}{R - RI)} = \frac{L}{R - \frac{La}{R}} = \frac{\frac{L}{R}}{1 - a\frac{L}{R}}$$

2. Plot the total delay as a function of $\frac{L}{R}$
   $x = \frac{L}{R}$

$$D(x) = \frac{x}{1 - ax} \quad \text{for} \quad 0 < x < \frac{1}{a}$$

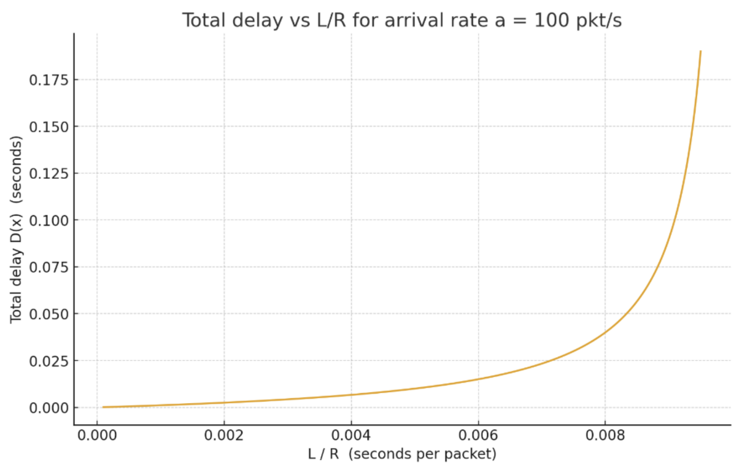

Figure 1:

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In modern packet-switched networks, including the Internet, the source host segments long, application-layer messages (for example, an image or a music file) into smaller packets and sends the packets into the network. The receiver then reassembles the packets back into the original message. We refer to this process as message segmentation. Figure 1.27 illustrates the end-to-end transport of a message with and without message segmentation. Consider a message that is 106 bits long that is to be sent from source to destination in Figure 1.27.

Suppose each link in the figure is 5 Mbps. Ignore propagation, queuing, and processing delays.

1. Consider sending the message from source to destination without message segmentation. How long does it take to move the message from the source host to the first packet switch? Keeping in mind that each switch uses store-and-forward packet switching, what is the total time to move the message from the source host to the destination host?
   Time to first switch: $\frac{10^6}{5*10^6} = 0.2s$
   Total end-to-end: $3 * 0.2s = 0.6s$

2. Now suppose that the message is segmented into 100 packets, with each packet being 10,000 bits long. How long does it take to move the first packet from the source host to the first switch? When the first packet is being sent from the first switch to the second switch, the second packet is being sent from the source host to the first switch. At what time will the second packet be fully received at the first switch?
   Per-packet time: $\frac{10^4}{5*10^6} = 0.002s = 2ms$
   First packet to first switch: $2ms$
   Second packet fully at first switch: $4ms$

3. How long does it take to move the file from the source host to the destination host when message segmentation is used? Compare this result with your answer in part (a) and comment.
   Store and forward pipeline with 3 links (k) and 100 packets (n)

   $$T = (n + k - 1)d = (100 + 3 - 1) * 2ms = 204ms$$

   204ms with segmentation vs 600ms without segmentation, almost 3x better performance

Chapter 2

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Consider Figure 2.12, for which there is an institutional network connected to the Internet. Suppose that the average object size is 1,000,000 bits and that the average request rate from the institution's browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average (see Section 2.2.5). Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to the institution router) and the average Internet delay. For the average access delay, use $\frac{\Delta}{1-\Delta b}$ where Δ is the average time required to send an object over the access link and b is the arrival rate of objects to the access link. Given: Average size: $L = 10^6$ bits.
Request rate: $b = 16$ req/s.

Internet delay: 3s.
Bit rate: $R = 15 * 10^6$
Access link rate: $\Delta = \frac{L}{R} = \frac{1}{15} = 0.0667s$

1. Find the total average response time.

$$\frac{\Delta}{1 - \Delta b} \Rightarrow \Delta b = \frac{1}{15} * 16 = \frac{16}{15} > 1$$

   $\Delta b > 1$ makes the system unstable therefore, the total average response time is unbounded because the queue grows unbounded.

2. Now, suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.4. Find the total response time.
   Only misses traverse the access link: $b' = 0.4 * 16 = 6.4$req/s.

$$\frac{\Delta}{1 - \Delta b'} = \frac{0.0667}{1 - 0.0667 * 6.4} = \frac{0.0667}{0.5731} \approx 0.116s$$

   $T = 0.4(3.0 + 0.116) = 1.2464s$

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1. What is a whois database?
   A public registry service mapping Internet resources (domain names, IP blocks, ASNs) to ownership/contact and registrar/route information.

2. Use various whois databases on the Internet to obtain the names of two DNS servers. Indicate which whois databases you used.

```
whois -h whois.educause.edu uoregon.edu | grep -A 5 'Name Servers\?'
    Name Servers:
     PHLOEM.UOREGON.EDU
     RUMINANT.UOREGON.EDU
     LSU-BDDS1.LSU.EDU

whois -h whois.verisign-grs.com google.com | grep -A 5 'Name Servers\?'
   Name Server: NS1.GOOGLE.COM
   Name Server: NS2.GOOGLE.COM
   Name Server: NS3.GOOGLE.COM
   Name Server: NS4.GOOGLE.COM
```

   I used EDUCAUSE database for .edu lookups and Verisign for .com lookups.

3. Use nslookup on your local host to send DNS queries to three DNS servers: your local DNS server and the two DNS servers you found in part (b). Try querying for Type A, NS, and MX reports. Summarize your findings.

```
nslookup -type=NS google.com NS1.GOOGLE.COM
    Server:  NS1.GOOGLE.COM
    Address: 2001:4860:4802:32::a#53

    google.com nameserver = ns2.google.com.
    google.com nameserver = ns3.google.com.
    google.com nameserver = ns1.google.com.
    google.com nameserver = ns4.google.com.

nslookup -type=A google.com NS1.GOOGLE.COM
    Server:  NS1.GOOGLE.COM
    Address: 2001:4860:4802:32::a#53

    Name: google.com
    Address: 142.250.73.142

nslookup -type=MX google.com NS1.GOOGLE.COM
    Server:  NS1.GOOGLE.COM
    Address: 2001:4860:4802:32::a#53

    google.com mail exchanger = 10 smtp.google.com.

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nslookup -type=NS uoregon.edu RUMINANT.UOREGON.EDU
    Server:  RUMINANT.UOREGON.EDU
    Address: 2001:468:d01:3c::80df:3c16#53

    uoregon.edu nameserver = lsu-bdds1.lsu.edu.
    uoregon.edu nameserver = ns1.f5cloudservices.com.
    uoregon.edu nameserver = phloem.uoregon.edu.
    uoregon.edu nameserver = ruminant.uoregon.edu.

nslookup -type=A uoregon.edu RUMINANT.UOREGON.EDU
    Server:  RUMINANT.UOREGON.EDU
    Address: 2001:468:d01:3c::80df:3c16#53

    Name: uoregon.edu
    Address: 23.185.0.3

nslookup -type=MX uoregon.edu RUMINANT.UOREGON.EDU
    Server:  RUMINANT.UOREGON.EDU
    Address: 2001:468:d01:3c::80df:3c16#53

    uoregon.edu mail exchanger = 10 mxb-000bfd01.gslb.pphosted.com.
    uoregon.edu mail exchanger = 10 mxa-000bfd01.gslb.pphosted.com.
```

```
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nslookup -type=NS uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    uoregon.edu nameserver = lsu-bdds1.lsu.edu.
    uoregon.edu nameserver = phloem.uoregon.edu.
    uoregon.edu nameserver = ns1.f5cloudservices.com.
    uoregon.edu nameserver = ruminant.uoregon.edu.

    Authoritative answers can be found from:
    phloem.uoregon.edu internet address = 128.223.32.35
    phloem.uoregon.edu has AAAA address 2001:468:d01:20::80df:2023
    ns1.f5cloudservices.com internet address = 107.162.234.197
    ruminant.uoregon.edu internet address = 128.223.60.22
    ruminant.uoregon.edu has AAAA address 2001:468:d01:3c::80df:3c16

nslookup -type=A uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    Name: uoregon.edu
    Address: 23.185.0.3

nslookup -type=MX uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    uoregon.edu mail exchanger = 10 mxb-000bfd01.gslb.pphosted.com.
    uoregon.edu mail exchanger = 10 mxa-000bfd01.gslb.pphosted.com.

    Authoritative answers can be found from:
```

google.com via NS1.GOOGLE.COM (authoritative): Returned the full NS set (ns1–ns4). A query returned one IPv4; MX returned smtp.google.com.

uoregon.edu via RUMINANT (authoritative): NS set included phloem, ruminant, lsu-bdds1 (secondary at LSU), and ns1.f5cloudservices.com (F5 managed). A returned 23.185.0.3 (Pantheon), MX pointed to Proofpoint (mxa/mxb-...pphosted.com).

Local resolver (recursive, non-authoritative): Matched the same NS/A/MX data, confirming cache consistency with authoritative sources.

4. Use nslookup to find a Web server that has multiple IP addresses. Does the Web server of your institution (school or company) have multiple IP addresses?

```
nslookup -type=A google.com
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    Name: google.com
    Address: 142.250.73.110

nslookup -type=AAAA google.com
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    google.com has AAAA address 2607:f8b0:400a:80a::200e

    Authoritative answers can be found from:

nslookup -type=A uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    Name: uoregon.edu
    Address: 23.185.0.3

nslookup -type=AAAA uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    uoregon.edu has AAAA address 2620:12a:8001::3
    uoregon.edu has AAAA address 2620:12a:8000::3

    Authoritative answers can be found from:

nslookup -type=A ruminant.uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53
```

```
    Non-authoritative answer:
    Name: ruminant.uoregon.edu
    Address: 128.223.60.22

nslookup -type=AAAA ruminant.uoregon.edu
    Server:  100.100.100.100
    Address: 100.100.100.100#53

    Non-authoritative answer:
    ruminant.uoregon.edu has AAAA address 2001:468:d01:3c::80df:3c16

    Authoritative answers can be found from:
```

uoregon.edu (the web presence) previously showed two AAAA records and one A (multiple IPs over IPv6).
ruminant.uoregon.edu (nameserver) resolved to one A and one AAAA (no multipath here).

5. Use the ARIN whois database to determine the IP address range used by your university.

```
whois -h whois.arin.net 128.223.60.22 | egrep -i 'OrgName|NetRange|CIDR'
    NetRange:       128.223.0.0 - 128.223.255.255
    CIDR:           128.223.0.0/16
    OrgName:        University of Oregon
```

6. Describe how an attacker can use whois databases and the nslookup tool to perform reconnaissance on an institution before launching an attack.
   (a) Identify ownership, admin contacts, and netblocks.
   (b) Enumerate authoritative DNS, subdomains: NS queries, zone walking if misconfigured, brute-force.
   (c) Map IP ranges/services to target infrastructure. Select phishing or network entry points.
7. Discuss why whois databases should be publicly available.
   Operational transparency, abuse handling, routing coordination, and incident response. Public data speeds troubleshooting and accountability, while modern policies mitigate privacy risks.